25-08-2026
24261A6625

Experiment 5:

AIM: Implementation of Multiple Linear Regression for House Price Prediction using sklearn

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn import metrics
USAhousing=pd.read_csv("C:/Users/MPIT/Downloads/USA_Housing.csv")
print(USAhousing.head())
X=USAhousing[['Avg. Area Income','Avg. Area House Age','Avg. Area Number of Rooms,
              'Avg. Area Number of Bedrooms','Area Population']]
y=USAhousing['Price']
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=0)
lm=LinearRegression()
lm.fit(X_train,y_train)
predictions=lm.predict(X_test)
plt.scatter(y_test,predictions)
plt.xlabel('Actual Price')
plt.ylabel('Predicted Prices')
plt.title('Actual vs Predicted Prices')
plt.show()
print('MAE(Mean Absolute Error):',metrics.mean_absolute_error(y_test,predictions))
print('MSE(Mean Squared Error):',metrics.mean_squared_error(y_test,predictions))
print('RMSE(Root Mean Squared Error):',np.sqrt(metrics.mean_squared_error(y_test,predictions))
```

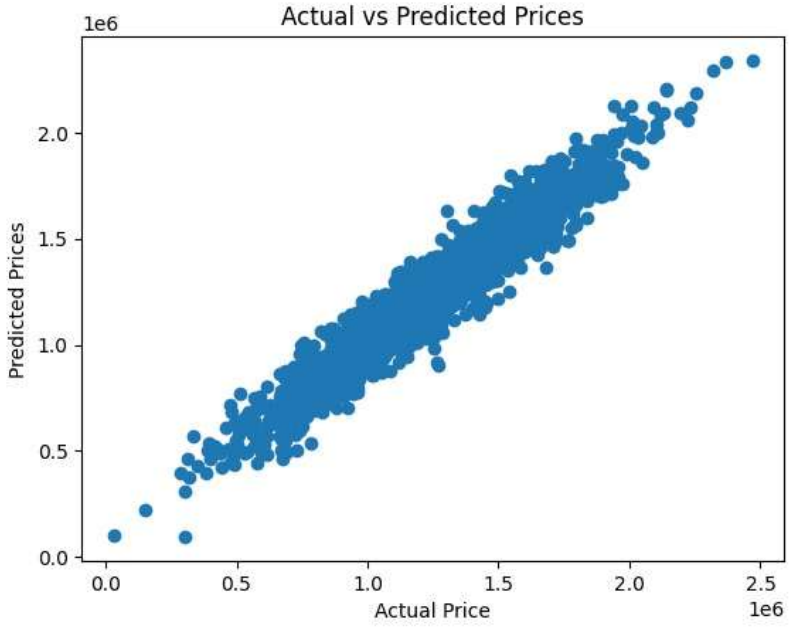
Output:

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms
0	79545.458574	5.682861	7.009188
1	79248.642455	6.002900	6.730821
2	61287.067179	5.865890	8.512727
3	63345.240046	7.188236	5.586729
4	59982.197226	5.040555	7.839388

	Avg. Area Number of Bedrooms	Area Population	Price
0	4.09	23086.800503	1.059034e+06
1	3.09	40173.072174	1.505891e+06
2	5.13	36882.159400	1.058988e+06
3	3.26	34310.242831	1.260617e+06
4	4.23	26354.109472	6.309435e+05

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	Address
0	208 chael Ferry Apt. 674\nLaurabury, NE 3701...
1	188 Johnson Views Suite 079\nLake Kathleen, CA...
2	9127 Elizabeth Stravenue\nDanieltown, WI 06482...
3	USS Barnett\nFPO AP 44820
4	USNS Raymond\nFPO AE 09386



MAE(Mean Absolute Error): 81563.14733992997
MSE(Mean Squared Error): 10337337828.264475
RMSE(Root Mean Absolute Error): 101672.6995228536